

EE 4170 Electromagnetic Radiation, Antennas, and Propagation

Rectangular Microstrip Patch Antenna with Coaxial Feeding at 10 GHz Resonance

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Introduction

In this project, we will be creating a rectangular microstrip patch antenna with a coaxial feed. We have a set list of requirements that we will get from example 14.1 from Antenna Theory by Balanis. The antenna needs to be able to be fabricated for practical applications.

Requirements

- S_{11} - parameter for the resonant frequency needs to be 10 GHz
- Use RT/duroid 5880 substrate
- Coaxial Feed with 50Ω impedance
- Plot input impedance both real and imaginary
- Add plots for the parametric sweep for the finalized dimensions
- Plot radiation patterns in both 3D and cuts
- Have an acceptable gain for practical implications

Background

A rectangular patch antenna is a widely used antenna configuration since it is easy to work with. There are many different types of rectangular patch antenna configurations such as feeding a microstrip, having a coaxial feed, adjusting resonance with an inset feed, stub, and many more different ways to affect how the patch is tuned. The basic configuration of this antenna will consist of a layer of conductive material such as copper or PEC that will make up the geometry of the antenna, a layer of substrate dielectric material, and finally a layer that covers the entire dielectric on the bottom as the ground made up of copper or PEC.

We will have different variables that will affect the geometry of our antenna. These will be shown in Table 1.

Name	Value (mm)	Description
t	0.035	Thickness of ground
h	1.6	Height of dielectric
W	9.4	Width of the patch
L	3.5	Length of the patch
Ltot	20	Total Length
Wtot	20	Total Width
R1	0.3175	Radius of the coaxial conductor
R2	1.025	Radius of the Teflon
Xf	0.75	X axis position of the coax
Yf	1	Y axis position of the coax
CoaxLength	-1.8625	Length of the coax feed

Table 1: Variables

Procedure

We first must define our variables in CST since we will be using them the entire time; we are going to make the geometry using them. We will start with creating the ground plane and that will be using the W_{tot} , L_{tot} , and t variables. We will then use the same variables for the dielectric brick however we will also include using h since that will basically be the thickness for our dielectric. Now to create the patch we will use the same thickness as the ground brick, but we will use L and W instead of L_{tot} and W_{tot} . To create the coax feed we will be starting with the rod that will connect to the patch and then use $R1$. To use create tefton we will make the wider diameter $R2$ and the inner $R1$ so that it surrounds the first cylinder. Finally, we have a .1mm gap that will shell the tefton cylinder and connect to the ground plane. For materials make sure to use PEC or Copper for the rod that will connect to the patch, the patch itself, the shell for the coax, and finally the ground plane. For the dielectric of the patch antenna make sure to use RT/duroid 5880 material. For the inner cylinder of the coax make sure to use PTFE to get the 50 ohm impedance matching.

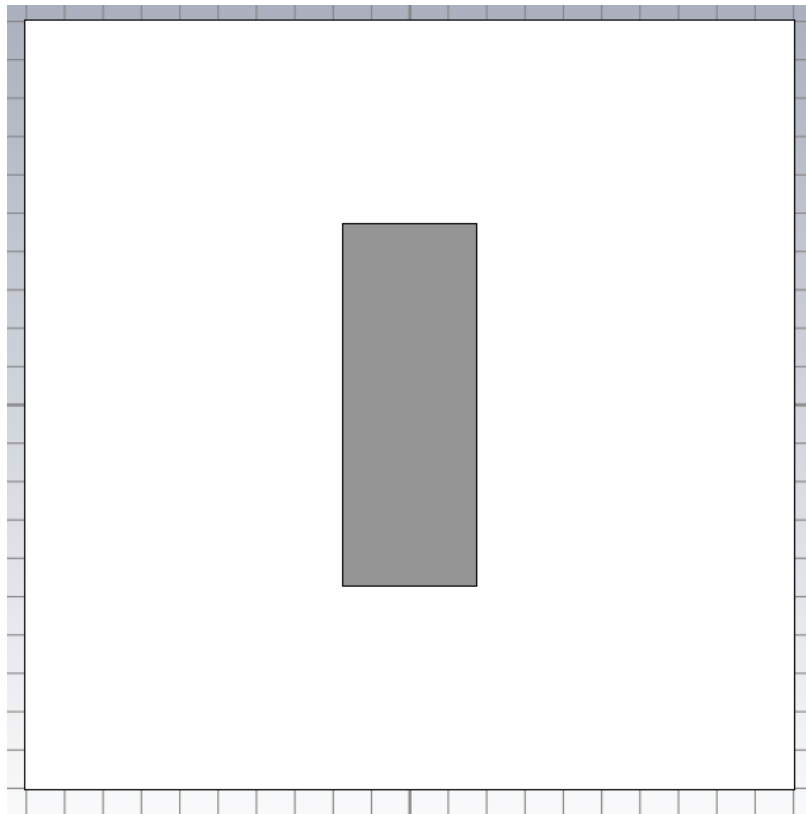


Figure 1: Patch antenna from above

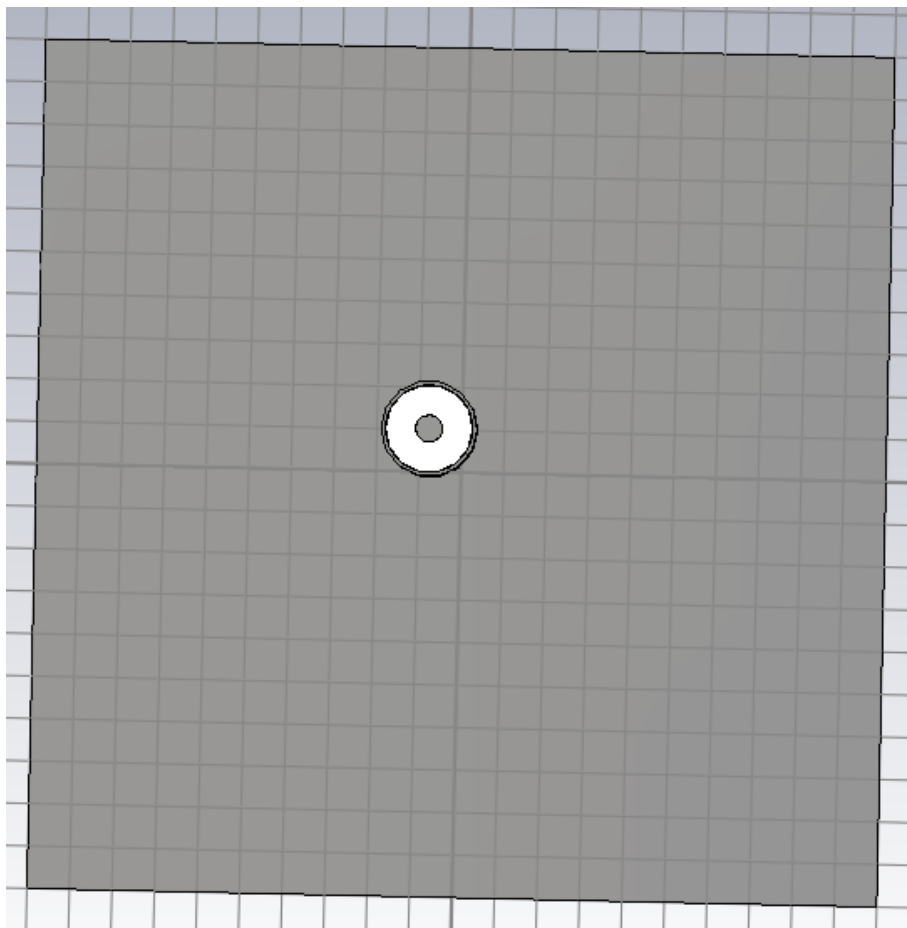


Figure 2: Patch antenna from below

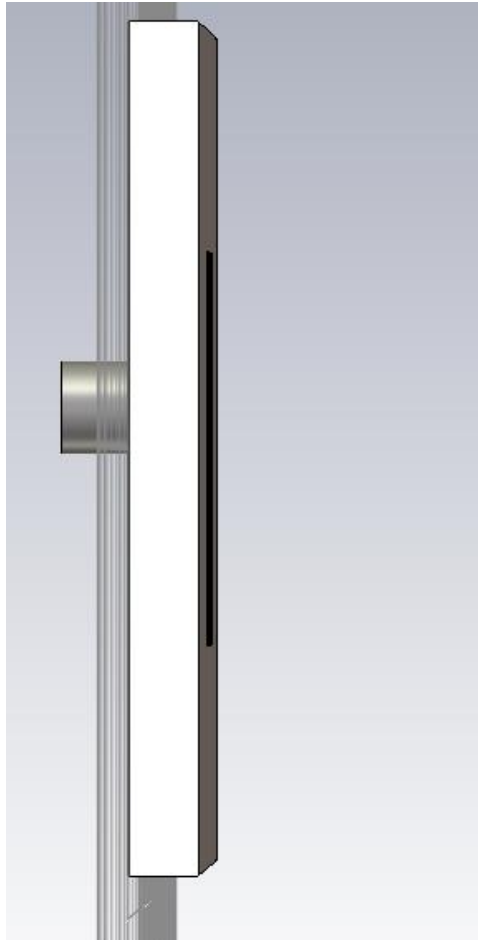


Figure 3: Patch antenna from the side

Analysis

We can see that we have acceptable resonance at 10 GHz since it is below 20 dB. We are also able to see that we have about 7dB gain. In the farfield region we able to see that it is in the +z direction.

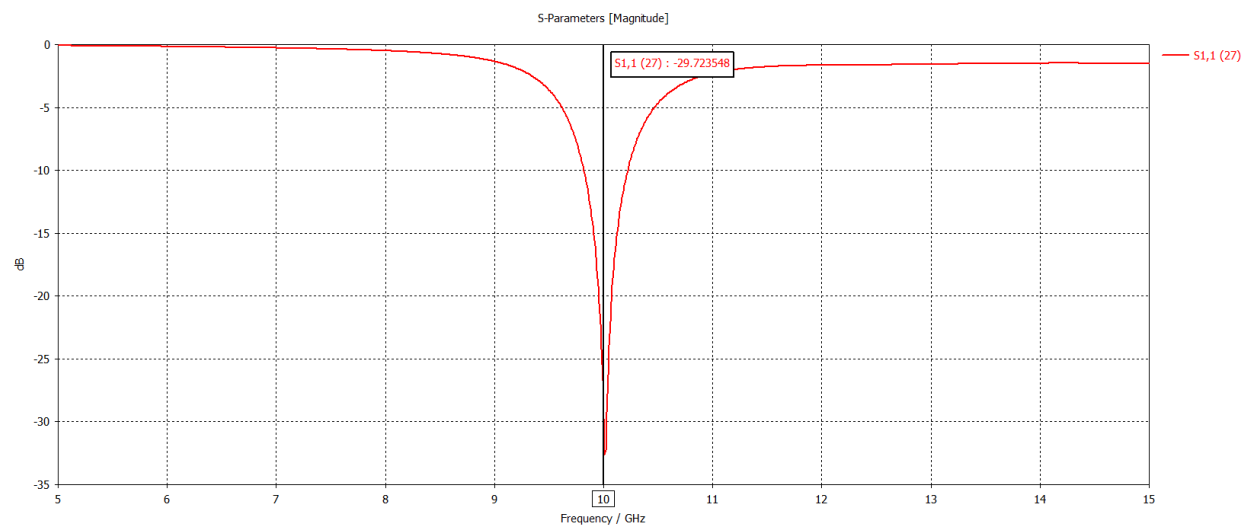


Figure 4: S11 Resonance graph at 10 GHz

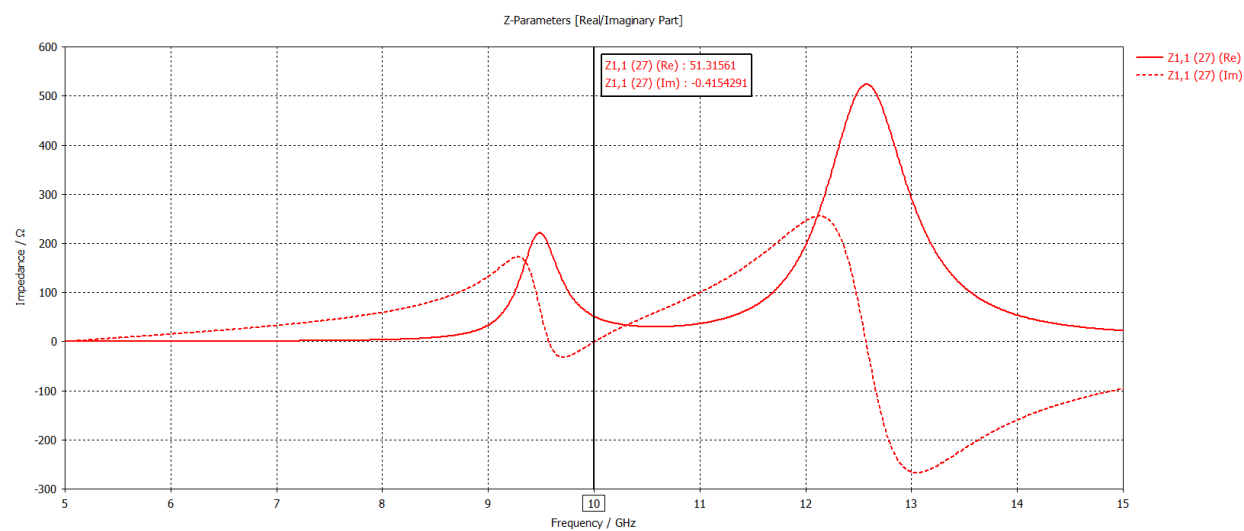


Figure 5: Impedance both imaginary and real at 10 GHz

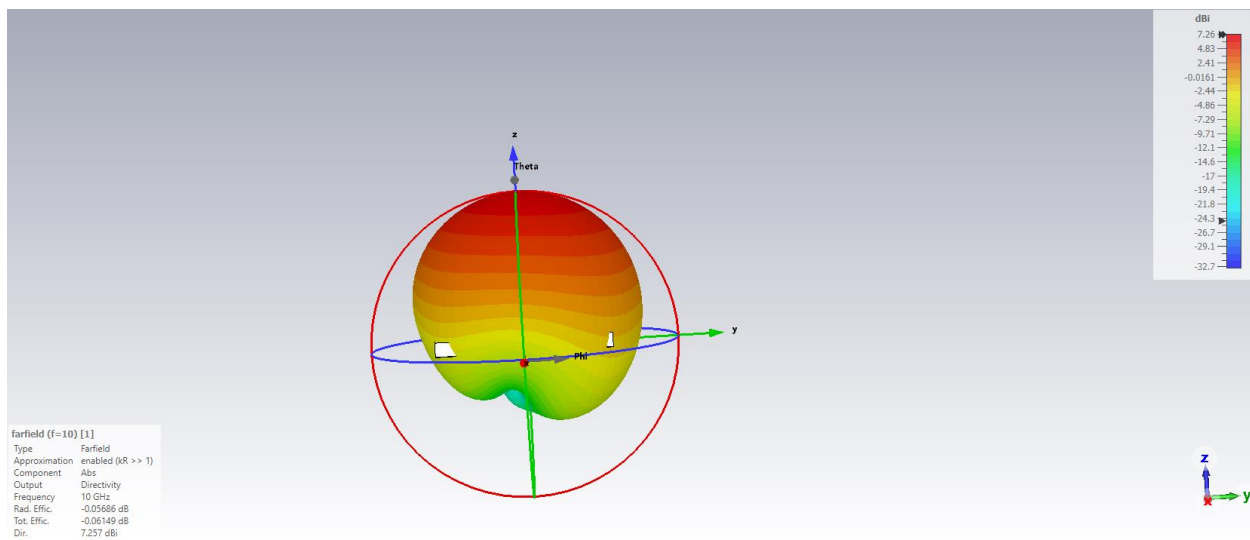


Figure 6: 3D Farfield

farfield (f=10) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Directivity
Frequency	10 GHz
Rad. Effic.	-0.05686 dB
Tot. Effic.	-0.06149 dB
Dir.	7.257 dBi

Figure 7: Directivity from 3D Farfield Plot

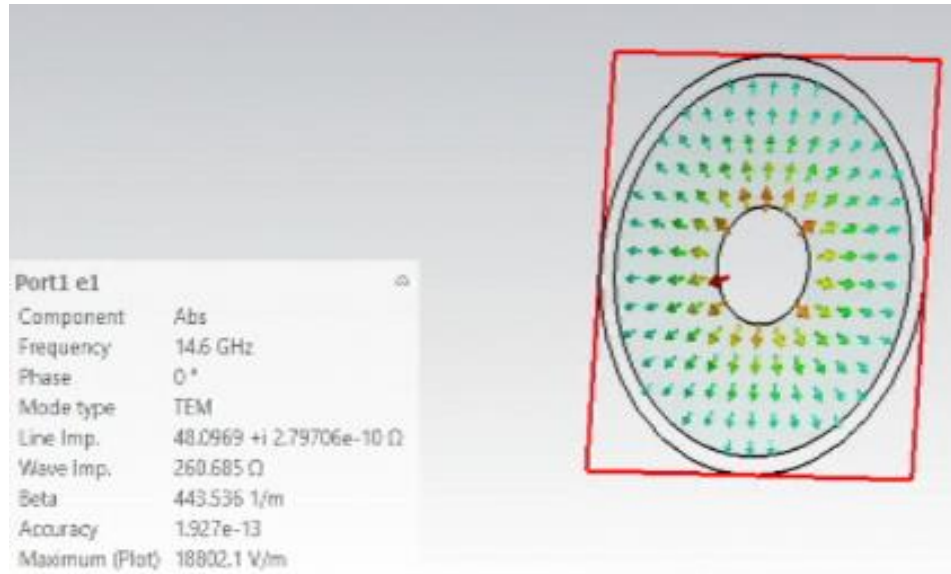


Figure 8: Port Impedance

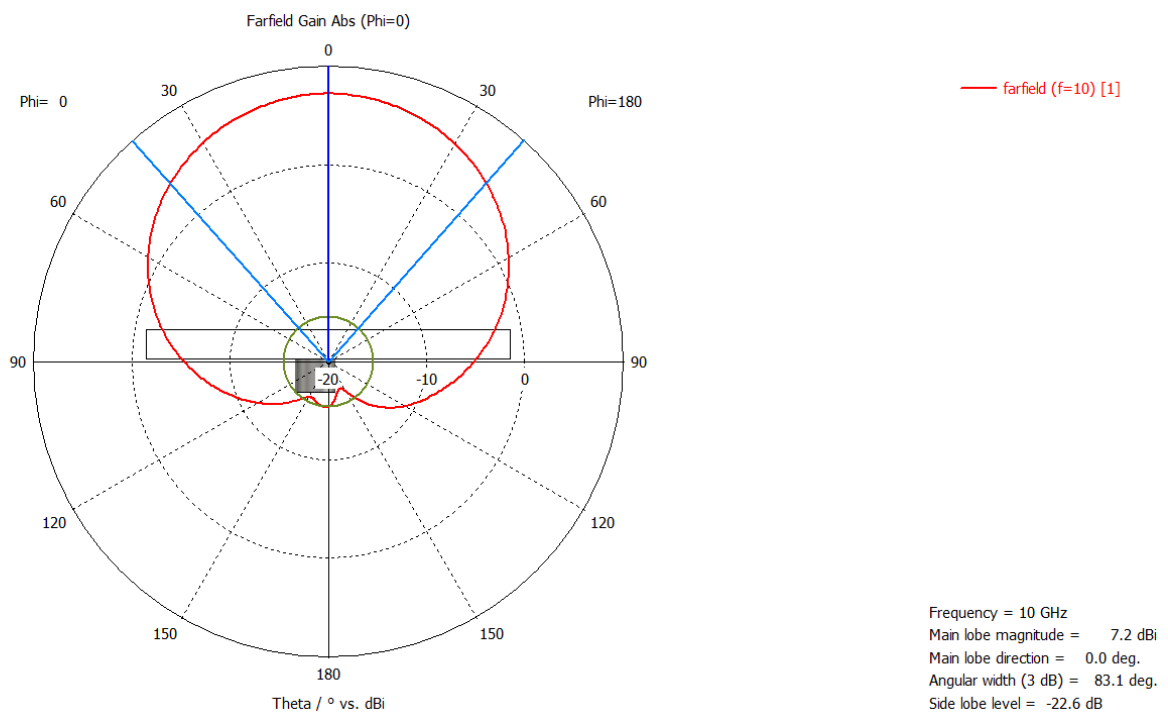


Figure 9: XZ Plane Farfield Cut

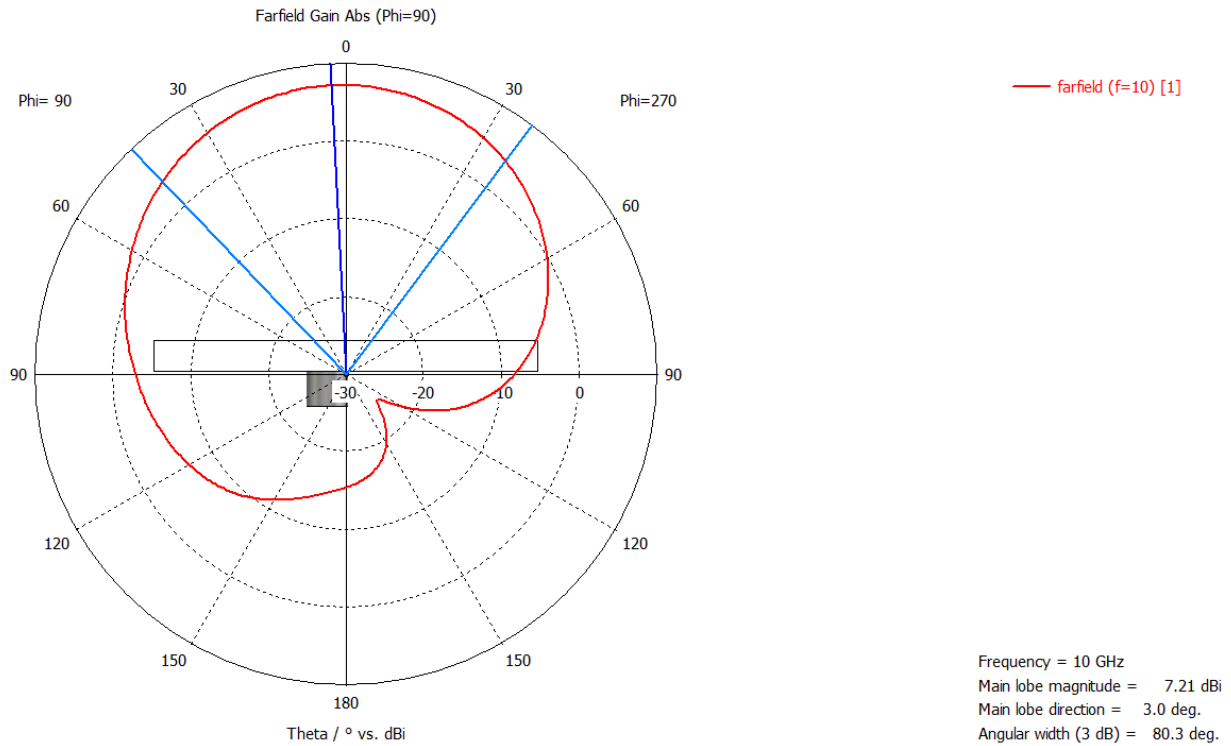


Figure 10: YZ Plane Farfield Cut

Conclusion

Overall, this antenna is able to be manufactured for use at 10GHz. The gain is at an acceptable level. This project was a trial of misunderstanding and being able to find the correct values for the antenna was a little bit of trial and error. I initially had made a design with the incorrect materials and did get resonance at 10 GHz. However, I did not meet the design requirement of using the proper material type which gave me impedance mismatches. I did end up correcting this and had to restart the project essentially. However, I got the proper materials, and this meant that it was able to be manufactured properly since the previous design was most likely not able to be manufactured.